

PrepareSampleDataFile

November 10, 2025

```
[3]: !pip install ydata-profiling

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.metrics import classification_report
from sklearn import metrics
from sklearn.metrics import accuracy_score
import json
from google.colab import drive
drive.mount('/content/drive') # Mount Google Drive to access files

# Function to load the JSON file line by line
def load_yelp_data(file_path):
    """
    Load Yelp JSON data file and convert it to a pandas DataFrame

    Args:
        file_path (str): Path to the JSON file

    Returns:
        pandas.DataFrame: DataFrame containing the loaded data
    """
    data = []
    with open(file_path, 'r') as f:
        for line in f:
            data.append(json.loads(line))
    return pd.DataFrame(data)
```

Collecting ydata-profiling

Downloading ydata_profiling-4.17.0-py2.py3-none-any.whl.metadata (22 kB)

Collecting scipy<1.16,>=1.4.1 (from ydata-profiling)

Downloading

scipy-1.15.3-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata
(61 kB)

62.0/62.0 kB

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Requirement already satisfied: pandas!=1.4.0,<3.0,>1.1 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (2.2.2)

Requirement already satisfied: matplotlib<=3.10,>=3.5 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (3.10.0)

Requirement already satisfied: pydantic>=2 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (2.11.10)

Requirement already satisfied: PyYAML<6.1,>=5.0.0 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (6.0.3)

Requirement already satisfied: jinja2<3.2,>=2.11.1 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (3.1.6)

Collecting visions<0.8.2,>=0.7.5 (from visions[type_image_path]<0.8.2,>=0.7.5->ydata-profiling)

Downloading visions-0.8.1-py3-none-any.whl.metadata (11 kB)

Requirement already satisfied: numpy<2.2,>=1.16.0 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (2.0.2)

Collecting minify-html>=0.15.0 (from ydata-profiling)

Downloading minify_html-0.18.1-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (18 kB)

Collecting filetype>=1.0.0 (from ydata-profiling)

Downloading filetype-1.2.0-py2.py3-none-any.whl.metadata (6.5 kB)

Collecting phik<0.13,>=0.11.1 (from ydata-profiling)

Downloading phik-0.12.5-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl.metadata (5.6 kB)

Requirement already satisfied: requests<3,>=2.24.0 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (2.32.4)

Requirement already satisfied: tqdm<5,>=4.48.2 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (4.67.1)

Requirement already satisfied: seaborn<0.14,>=0.10.1 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (0.13.2)

Collecting multimethod<2,>=1.4 (from ydata-profiling)

Downloading multimethod-1.12-py3-none-any.whl.metadata (9.6 kB)

Requirement already satisfied: statsmodels<1,>=0.13.2 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (0.14.5)

Requirement already satisfied: typeguard<5,>=3 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (4.4.4)

Collecting imagehash==4.3.1 (from ydata-profiling)

Downloading ImageHash-4.3.1-py2.py3-none-any.whl.metadata (8.0 kB)

Requirement already satisfied: wordcloud>=1.9.3 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (1.9.4)

Collecting dacite>=1.8 (from ydata-profiling)

Downloading dacite-1.9.2-py3-none-any.whl.metadata (17 kB)

Requirement already satisfied: numba<=0.61,>=0.56.0 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (0.60.0)

Requirement already satisfied: PyWavelets in /usr/local/lib/python3.12/dist-packages (from imagehash==4.3.1->ydata-profiling) (1.9.0)

Requirement already satisfied: pillow in /usr/local/lib/python3.12/dist-packages (from imagehash==4.3.1->ydata-profiling) (11.3.0)

Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2<3.2,>=2.11.1->ydata-profiling) (3.0.3)

Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib<=3.10,>=3.5->ydata-profiling) (1.3.3)

Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib<=3.10,>=3.5->ydata-profiling) (0.12.1)

Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib<=3.10,>=3.5->ydata-profiling) (4.60.1)

Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib<=3.10,>=3.5->ydata-profiling) (1.4.9)

Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib<=3.10,>=3.5->ydata-profiling) (25.0)

Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib<=3.10,>=3.5->ydata-profiling) (3.2.5)

Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib<=3.10,>=3.5->ydata-profiling) (2.9.0.post0)

Requirement already satisfied: llvmlite<0.44,>=0.43.0dev0 in /usr/local/lib/python3.12/dist-packages (from numba<=0.61,>=0.56.0->ydata-profiling) (0.43.0)

Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas!=1.4.0,<3.0,>1.1->ydata-profiling) (2025.2)

Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas!=1.4.0,<3.0,>1.1->ydata-profiling) (2025.2)

Requirement already satisfied: joblib>=0.14.1 in /usr/local/lib/python3.12/dist-packages (from phik<0.13,>=0.11.1->ydata-profiling) (1.5.2)

Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic>=2->ydata-profiling) (0.7.0)

Requirement already satisfied: pydantic-core==2.33.2 in /usr/local/lib/python3.12/dist-packages (from pydantic>=2->ydata-profiling) (2.33.2)

Requirement already satisfied: typing-extensions>=4.12.2 in /usr/local/lib/python3.12/dist-packages (from pydantic>=2->ydata-profiling) (4.15.0)

Requirement already satisfied: typing-inspection>=0.4.0 in /usr/local/lib/python3.12/dist-packages (from pydantic>=2->ydata-profiling) (0.4.2)

Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests<3,>=2.24.0->ydata-profiling) (3.4.4)

Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-

packages (from requests<3,>=2.24.0->ydata-profiling) (3.11)
 Requirement already satisfied: urllib3<3,>=1.21.1 in
 /usr/local/lib/python3.12/dist-packages (from requests<3,>=2.24.0->ydata-
 profiling) (2.5.0)
 Requirement already satisfied: certifi>=2017.4.17 in
 /usr/local/lib/python3.12/dist-packages (from requests<3,>=2.24.0->ydata-
 profiling) (2025.10.5)
 Requirement already satisfied: patsy>=0.5.6 in /usr/local/lib/python3.12/dist-
 packages (from statsmodels<1,>=0.13.2->ydata-profiling) (1.0.2)
 Requirement already satisfied: attrs>=19.3.0 in /usr/local/lib/python3.12/dist-
 packages (from
 visions<0.8.2,>=0.7.5->visions[type_image_path]<0.8.2,>=0.7.5->ydata-profiling)
 (25.4.0)
 Requirement already satisfied: networkx>=2.4 in /usr/local/lib/python3.12/dist-
 packages (from
 visions<0.8.2,>=0.7.5->visions[type_image_path]<0.8.2,>=0.7.5->ydata-profiling)
 (3.5)
 Collecting puremagic (from
 visions<0.8.2,>=0.7.5->visions[type_image_path]<0.8.2,>=0.7.5->ydata-profiling)
 Downloading puremagic-1.30-py3-none-any.whl.metadata (5.8 kB)
 Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-
 packages (from python-dateutil>=2.7->matplotlib<=3.10,>=3.5->ydata-profiling)
 (1.17.0)
 Downloading ydata_profiling-4.17.0-py2.py3-none-any.whl (399 kB)
 399.3/399.3 kB
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 Downloading ImageHash-4.3.1-py2.py3-none-any.whl (296 kB)
 296.5/296.5 kB
 20.1 MB/s eta 0:00:00
 Downloading dacite-1.9.2-py3-none-any.whl (16 kB)
 Downloading filetype-1.2.0-py2.py3-none-any.whl (19 kB)
 Downloading
 minify_html-0.18.1-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl
 (3.1 MB)
 3.1/3.1 MB
 42.0 MB/s eta 0:00:00
 Downloading multimethod-1.12-py3-none-any.whl (10 kB)
 Downloading
 phik-0.12.5-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl (679 kB)
 679.7/679.7 kB
 37.8 MB/s eta 0:00:00
 Downloading
 scipy-1.15.3-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (37.3
 MB)
 37.3/37.3 MB
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 Downloading visions-0.8.1-py3-none-any.whl (105 kB)
 105.4/105.4 kB

7.5 MB/s eta 0:00:00

Downloading puremagic-1.30-py3-none-any.whl (43 kB)

43.3/43.3 kB

2.8 MB/s eta 0:00:00

Installing collected packages: puremagic, minify-html, filetype, scipy, multimethod, dacite, imagehash, visions, phik, ydata-profiling

Attempting uninstall: scipy

Found existing installation: scipy 1.16.3

Uninstalling scipy-1.16.3:

Successfully uninstalled scipy-1.16.3

Successfully installed dacite-1.9.2 filetype-1.2.0 imagehash-4.3.1 minify-html-0.18.1 multimethod-1.12 phik-0.12.5 puremagic-1.30 scipy-1.15.3 visions-0.8.1 ydata-profiling-4.17.0

Mounted at /content/drive

```
[5]: from google.colab import drive
drive.mount('/content/drive')
file_path = '/content/drive/My Drive/notebooks/Yelp JSON/yelp_dataset (Unzipped_
↳Files)/yelp_academic_dataset_user.json'
chunksize = 100000 # Process data in chunks of 100,000 rows
review_count = 10 # Filter criterion: minimum number of reviews
filtered_users = []

# Process JSON file in chunks to handle large dataset
for chunk in pd.read_json(file_path, lines=True, chunksize=chunksize):
    # Filter by business_id
    filtered = chunk[chunk['review_count']>=review_count]
    filtered_users.append(filtered)

# # Combine all filtered chunks into single DataFrame
filtered_users_df = pd.concat(filtered_users, ignore_index=True)

# Save filtered dataset to a file, temporal storage
filtered_users_df.to_csv('/content/drive/My Drive/notebooks/Yelp JSON/
↳yelp_dataset (Unzipped Files)/yelp_filtered_users_with_more_than_n_reviews.
↳csv', index=False)

print(f"Filtered dataset shape: {filtered_users_df.shape}")
print(filtered_users_df.head())
print(f"Filtered dataset descriptive statistics: {filtered_users_df.describe()}")
```

Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).

Filtered dataset shape: (726519, 22)

	user_id	name	review_count	yelping_since	useful	\
0	qVc80DYU5SZjKXVBgXdI7w	Walker	585	2007-01-25 16:47:26	7217	
1	j14WgRoU_-2ZE1aw1dXrJg	Daniel	4333	2009-01-25 04:35:42	43091	
2	2WnXYQFK0hXEoTxPtV2zvg	Steph	665	2008-07-25 10:41:00	2086	

3	SZDeASXq7o05mMNLshsdIA	Gwen	224	2005-11-29 04:38:33	512
4	hA5lMy-EnncsH4JoR-hFGQ	Karen	79	2007-01-05 19:40:59	29

	funny	cool		elite \
0	1259	5994		2007
1	13066	27281	2009,2010,2011,2012,2013,2014,2015,2016,2017,2...	
2	1010	1003	2009,2010,2011,2012,2013	
3	330	299	2009,2010,2011	
4	15	7		

		friends	fans	...	\
0	NSCy54eWehBJyZdG2iE84w, pe42u7DcCH2QmI81NX-8qA...	267	...		
1	ueRPEOCX75ePGMq0FVj6IQ, 52oH4DrRvzzl8wh5UXyU0A...	3138	...		
2	Lu03Bn4f3rlhyHIaNFtlnA, j9B4XdHUhDfTKVecyWQgyA...	52	...		
3	enx1vVPnfdNUdPho6PH_wg, 4w0cvMLtU6a9Lslggq74Vg...	28	...		
4	PBK4q9KEEBHhFvSXCUIrIw, 3FWPpM7KU1gXeOM_ZbYmbA...	1	...		

	compliment_more	compliment_profile	compliment_cute	compliment_list \
0	65	55	56	18
1	264	184	157	251
2	13	10	17	3
3	4	1	6	2
4	1	0	0	0

	compliment_note	compliment_plain	compliment_cool	compliment_funny \
0	232	844	467	467
1	1847	7054	3131	3131
2	66	96	119	119
3	12	16	26	26
4	1	1	0	0

	compliment_writer	compliment_photos
0	239	180
1	1521	1946
2	35	18
3	10	9
4	0	0

[5 rows x 22 columns]

Filtered dataset descriptive statistics:				review_count	useful
funny	cool	\			
count	726519.000000	726519.000000	726519.000000	726519.000000	
mean	58.099403	111.435644	45.596217	64.271497	
std	129.404957	1057.515871	673.601985	933.786964	
min	10.000000	0.000000	0.000000	0.000000	
25%	15.000000	8.000000	1.000000	2.000000	
50%	24.000000	20.000000	4.000000	5.000000	
75%	52.000000	52.000000	12.000000	14.000000	

max	17473.000000	206296.000000	185823.000000	199878.000000
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	fans	average_stars	compliment_hot	compliment_more \
count	726519.000000	726519.000000	726519.000000	726519.000000
mean	3.860708	3.823840	4.920835	0.778748
std	29.807858	0.665118	121.684542	21.204413
min	0.000000	1.000000	0.000000	0.000000
25%	0.000000	3.460000	0.000000	0.000000
50%	0.000000	3.910000	0.000000	0.000000
75%	2.000000	4.290000	0.000000	0.000000
max	12497.000000	5.000000	25784.000000	13501.000000

	compliment_profile	compliment_cute	compliment_list	compliment_note \
count	726519.000000	726519.000000	726519.000000	726519.000000
mean	0.487556	0.362982	0.174576	3.869822
std	25.065912	18.783043	16.612999	99.478736
min	0.000000	0.000000	0.000000	0.000000
25%	0.000000	0.000000	0.000000	0.000000
50%	0.000000	0.000000	0.000000	0.000000
75%	0.000000	0.000000	0.000000	1.000000
max	14180.000000	13654.000000	12669.000000	59031.000000

	compliment_plain	compliment_cool	compliment_funny	compliment_writer \
count	726519.000000	726519.000000	726519.000000	726519.000000
mean	8.14752	7.696258	7.696258	2.862441
std	197.38126	159.778293	159.778293	53.180044
min	0.000000	0.000000	0.000000	0.000000
25%	0.000000	0.000000	0.000000	0.000000
50%	0.000000	0.000000	0.000000	0.000000
75%	1.000000	1.000000	1.000000	0.000000
max	101097.000000	49967.000000	49967.000000	15934.000000

	compliment_photos
count	726519.000000
mean	3.344953
std	157.381638
min	0.000000
25%	0.000000
50%	0.000000
75%	0.000000
max	82630.000000

```
[6]: # load business data file
file_path = '/content/drive/My Drive/notebooks/Yelp JSON/yelp_dataset (Unzipped,
Files)/yelp_academic_dataset_business.json'
df_business = load_yelp_data(file_path)
```

```

# Identify number of records
print(df_business.shape)

# describe data types
print(df_business.describe())

print(df_business.info())

# Filter out businesses that has less than 20 reviews
df_business_filtered = df_business[(df_business['categories'].str.
    ↪contains('Restaurants', na=False)) & (df_business['review_count'] >= 20)]

print(df_business_filtered.head())
print(df_business_filtered.shape)
print(df_business_filtered.describe())
print(df_business_filtered.info())

```

```
(150346, 14)
```

	latitude	longitude	stars	review_count \
count	150346.000000	150346.000000	150346.000000	150346.000000
mean	36.671150	-89.357339	3.596724	44.866561
std	5.872759	14.918502	0.974421	121.120136
min	27.555127	-120.095137	1.000000	5.000000
25%	32.187293	-90.357810	3.000000	8.000000
50%	38.777413	-86.121179	3.500000	15.000000
75%	39.954036	-75.421542	4.500000	37.000000
max	53.679197	-73.200457	5.000000	7568.000000

	is_open
count	150346.00000
mean	0.79615
std	0.40286
min	0.00000
25%	1.00000
50%	1.00000
75%	1.00000
max	1.00000

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 150346 entries, 0 to 150345
```

```
Data columns (total 14 columns):
```

#	Column	Non-Null Count	Dtype
0	business_id	150346 non-null	object
1	name	150346 non-null	object
2	address	150346 non-null	object
3	city	150346 non-null	object
4	state	150346 non-null	object


```

5  postal_code    150346 non-null  object
6  latitude       150346 non-null  float64
7  longitude      150346 non-null  float64
8  stars          150346 non-null  float64
9  review_count   150346 non-null  int64
10 is_open        150346 non-null  int64
11 attributes     136602 non-null  object
12 categories     150243 non-null  object
13 hours          127123 non-null  object

```

dtypes: float64(3), int64(2), object(9)

memory usage: 16.1+ MB

None

	business_id	name	address	city \
3	MTSW4McQd7CbVtyjqoe9mw	St Honore Pastries	935 Race St	Philadelphia
12	il_Ro8jwPlHresjw9EGmBg	Denny's	8901 US 31 S	Indianapolis
14	ObPLkLQqhHP05kt1_EXmNQ	Zio's Italian Market	2575 E Bay Dr	Largo
15	MUTTqe8uqyMdB1186RmNeA	Tuna Bar	205 Race St	Philadelphia
19	ROeacJQwBeh05Rqg7F6TCg	BAP	1224 South St	Philadelphia

	state	postal_code	latitude	longitude	stars	review_count	is_open \
3	PA	19107	39.955505	-75.155564	4.0	80	1
12	IN	46227	39.637133	-86.127217	2.5	28	1
14	FL	33771	27.916116	-82.760461	4.5	100	0
15	PA	19106	39.953949	-75.143226	4.0	245	1
19	PA	19147	39.943223	-75.162568	4.5	205	1

	attributes \
3	{'RestaurantsDelivery': 'False', 'OutdoorSeati...
12	{'RestaurantsReservations': 'False', 'Restaura...
14	{'OutdoorSeating': 'False', 'RestaurantsGoodFo...
15	{'RestaurantsReservations': 'True', 'Restauran...
19	{'NoiseLevel': 'u'quiet'', 'GoodForMeal': '{'d...

	categories \
3	Restaurants, Food, Bubble Tea, Coffee & Tea, B...
12	American (Traditional), Restaurants, Diners, B...
14	Food, Delis, Italian, Bakeries, Restaurants
15	Sushi Bars, Restaurants, Japanese
19	Korean, Restaurants

	hours
3	{'Monday': '7:0-20:0', 'Tuesday': '7:0-20:0', ...
12	{'Monday': '6:0-22:0', 'Tuesday': '6:0-22:0', ...
14	{'Monday': '10:0-18:0', 'Tuesday': '10:0-20:0'...
15	{'Tuesday': '13:30-22:0', 'Wednesday': '13:30-...
19	{'Monday': '11:30-20:30', 'Tuesday': '11:30-20...

(33941, 14)

latitude	longitude	stars	review_count	is_open
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```

count    33941.000000    33941.000000    33941.000000    33941.000000    33941.000000
mean      36.595727     -87.978644      3.610383     128.627412      0.716744
std        5.676611      13.916103      0.741008     223.798599      0.450586
min       27.564457    -120.083748      1.000000      20.000000      0.000000
25%       32.132899     -90.230599      3.000000      35.000000      0.000000
50%       38.775632     -86.044090      3.500000      65.000000      1.000000
75%       39.952355     -75.358845      4.000000     138.000000      1.000000
max       53.649743     -74.664459      5.000000     7568.000000      1.000000

```

```
<class 'pandas.core.frame.DataFrame'>
```

```
Index: 33941 entries, 3 to 150336
```

```
Data columns (total 14 columns):
```

#	Column	Non-Null Count	Dtype
0	business_id	33941 non-null	object
1	name	33941 non-null	object
2	address	33941 non-null	object
3	city	33941 non-null	object
4	state	33941 non-null	object
5	postal_code	33941 non-null	object
6	latitude	33941 non-null	float64
7	longitude	33941 non-null	float64
8	stars	33941 non-null	float64
9	review_count	33941 non-null	int64
10	is_open	33941 non-null	int64
11	attributes	33911 non-null	object
12	categories	33941 non-null	object
13	hours	31762 non-null	object

```
dtypes: float64(3), int64(2), object(9)
```

```
memory usage: 3.9+ MB
```

```
None
```

```

[7]: # Run import pandas as pd
      from scipy.stats import shapiro

      # Extract the star ratings
      stars = df_business_filtered['stars'].dropna()

      # Perform Shapiro-Wilk test
      stat, p_value = shapiro(stars)

      print("Shapiro-Wilk Test statistic:", stat)
      print("p-value:", p_value)

      # Interpretation
      if p_value > 0.05:
          print("Fail to reject H0: Star ratings appear normally distributed.")
      else:

```

```

print("Reject H0: Star ratings are not normally distributed.")

# Visualized star rating

import matplotlib.pyplot as plt
import seaborn as sns

sns.histplot(stars, kde=True)
plt.title("Yelp Business Star Ratings distribution")
plt.xlabel("Stars")
plt.ylabel("Frequency")
plt.show()

```

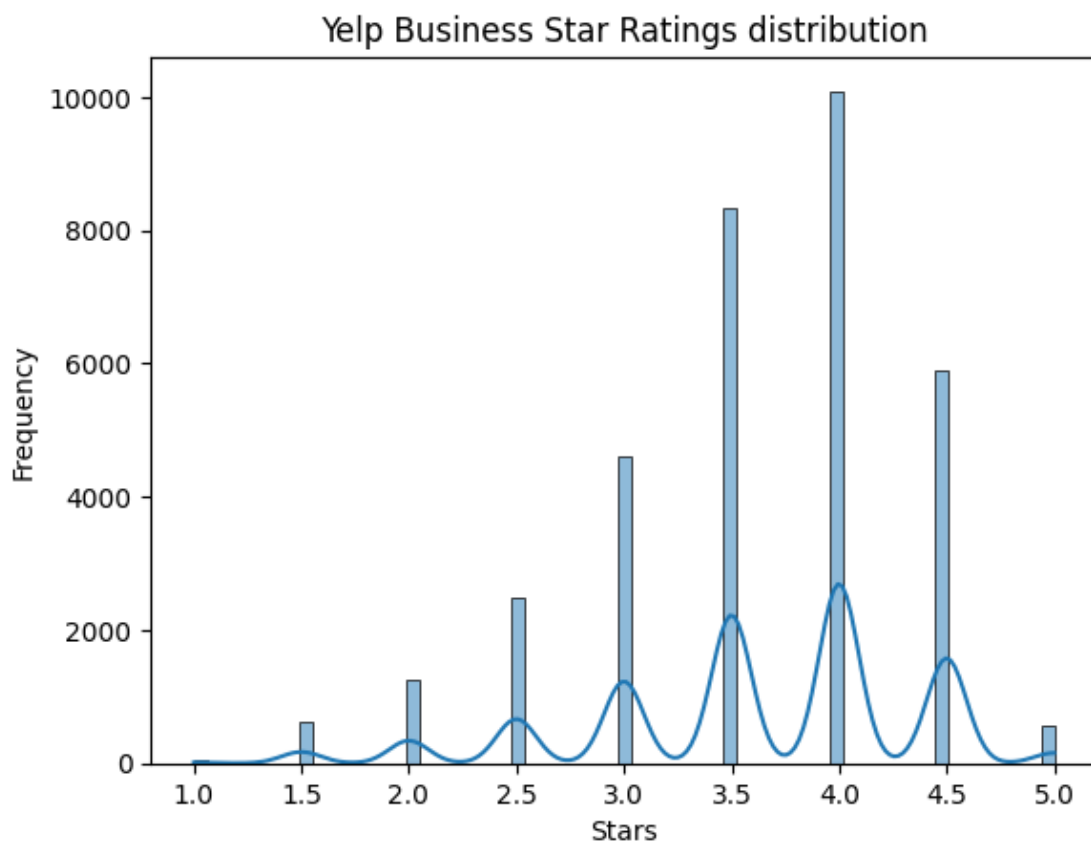
/usr/local/lib/python3.12/dist-packages/scipy/stats/_axis_nan_policy.py:586:
UserWarning: scipy.stats.shapiro: For N > 5000, computed p-value may not be
accurate. Current N is 33941.

```
res = hypotest_fun_out(*samples, **kwargs)
```

Shapiro-Wilk Test statistic: 0.9208025121724646

p-value: 7.299859920468648e-83

Reject H0: Star ratings are not normally distributed.



```
[8]: n_bins = 5
col = 'stars'
# Compute edges - from 0% to 100% quantiles (devide data into 5 equal parts)
quantile_edges = stars.quantile([i / n_bins for i in range(n_bins + 1)]).values
print("Quantile edges:", quantile_edges)
```

Quantile edges: [1. 3. 3.5 4. 4. 5.]

```
[9]: import pandas as pd
file_path = '/content/drive/My Drive/notebooks/Yelp JSON/yelp_dataset (Unzipped_
↳Files)/yelp_academic_dataset_review.json'

samples_per_star = 2000
chunksize = 1000000
filtered_reviews = []
sample = []
binned_chunks = []
target_sample = 100000 # Target 100k samples
sampling_frac = target_sample / 7_000_00 # whole dataset

for chunk in pd.read_json(file_path, lines=True, chunksize=100000):
    chunk['date'] = pd.to_datetime(chunk['date'], errors='coerce')

    # Date-based filtering (pre-COVID/post-COVID)
    filtered_chunk_1 = chunk[chunk['user_id'].isin(filtered_users_df['user_id'])
    & (chunk['date'] <='2020-03-11' )]
    filtered_chunk_2 = chunk[chunk['user_id'].isin(filtered_users_df['user_id'])
    & (chunk['date'] >= '2022-10-01' )]
    filtered_chunk = pd.concat([filtered_chunk_1, filtered_chunk_2])
    sampled_chunk = filtered_chunk.sample(frac=sampling_frac, random_state=42)

    # merge filtered chunk with business data and add star rating
    filtered_chunk = pd.merge(sampled_chunk,
    ↳df_business_filtered[['business_id', 'stars']], on='business_id',
    ↳how='left', suffixes=('', '_business'))
    # Bin star ratings using quantile-based edges
    filtered_chunk["stars_binned"] = pd.cut(
        filtered_chunk["stars_business"],
        bins=quantile_edges,
        labels=[1, 2, 3, 4],
        duplicates='drop'
    )

    binned_chunks.append(filtered_chunk)

# Combine all chunks into one final DataFrame
```

```
df_review_data_binned = pd.concat(binned_chunks, ignore_index=True)
df_review_data_binned.info()
df_review_data_binned.head()
df_review_data_binned.count
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 604965 entries, 0 to 604964
Data columns (total 11 columns):
#   Column                Non-Null Count  Dtype
---  -
0   review_id             604965 non-null object
1   user_id               604965 non-null object
2   business_id           604965 non-null object
3   stars                 604965 non-null int64
4   useful                604965 non-null int64
5   funny                 604965 non-null int64
6   cool                  604965 non-null int64
7   text                  604965 non-null object
8   date                  604965 non-null datetime64[ns]
9   stars_business        411681 non-null float64
10  stars_binned           411589 non-null category
dtypes: category(1), datetime64[ns](1), float64(1), int64(4), object(4)
memory usage: 46.7+ MB
```

```
[9]: <bound method DataFrame.count of                                review_id
user_id \
0      UQLoY-n1QT7roasrWBuudg  vfy68LKvXM7wEHoiUL-l0w
1      Sn6b49aTOHj_H4qlKA4qwQ  uuJAaJdA_3Dt4CPd-xSU7A
2      J-u53_Qt1Rsc1KeOK-udBQ  WY7B-CBeLbjjDBr7314k-A
3      aQWfMqoufuv4PEPakZDupQ  yqy_LSM4bGpVK7rmfGE8EA
4      GJsPnOd_JgPfMvBF4Xe1Vg  i_40kgr133EgKYVFdBSoZA
...
604960  LNyNv8mhTCCEXYRi-How5g  qZuyjPXIsRjyFBr7HNdVcw
604961  _owETWNyzeE5vDvGXIUfrag  yohyhEr_oXDjQaAzSxFf3Q
604962  DKtFLjR6EW05F8DnuZTTqg  It9h4fvS9BJxo5bmXbK-Vg
604963  SZmqr2AprQM1jVdyihduCw  W6nMsnSkXCuUsfq4ve_syg
604964  WVS5z0hrIe6w2z6pudfsNw  i_s_B01CFoZ3bw19vgsBgw

      business_id  stars  useful  funny  cool  \
0      _3m2qyCqx2pnDNyo5vYM3g      4      0      0      0
1      dyVKL3qKsz_XDxtjpbNcgQ      5      0      0      0
2      9nkYaCQRy00fi8SrXDFhGw      1      2      4      1
3      Ps-bfgV3aPvBx9UKdwCplA      5      3      0      1
4      8QJTzIaR5FceGds8HBSIig      5      0      0      0
...
604960  rgeuy1qbw6Z8B6CSVANHIA      5      3      1      2
604961  Meww6RIE4cFL_kGOaRlbAA      5      6      2      4
```

604962	QHWYlmVbLC3K6eglWoHVvA	4	1	0	0
604963	5zTGhEeKl6Kz-ULDgy0urg	5	5	2	5
604964	raBxnVejZrpQT4dfwABTEg	1	1	0	0

	text	date	\
0	Food is always good, with interesting specials...	2015-08-09 14:35:24	
1	Excellent service, delicious food and the atmo...	2014-01-02 04:25:46	
2	This place! Let's see...everything here is 1...	2015-02-16 21:32:30	
3	Love Hope Flowers after years and years buying...	2017-05-13 18:53:46	
4	The Barrio IPA is awesome! Paired with beer sp...	2013-05-29 13:48:24	
...	...	...	
604960	Loved the strawberry lavender donut! Got my do...	2017-06-14 03:59:28	
604961	I've been looking for a go-to BBQ place in New...	2019-07-04 21:46:07	
604962	This is the place I bring friends and family w...	2020-02-14 02:41:56	
604963	A beer lovers trip to Philly is not complete w...	2018-04-18 19:10:34	
604964	I was called 5mins after my appt with a resche...	2013-11-12 21:18:28	

	stars_business	stars_binned
0	4.0	3
1	3.5	2
2	NaN	NaN
3	NaN	NaN
4	4.0	3
...	...	...
604960	4.0	3
604961	4.5	4
604962	4.0	3
604963	4.0	3
604964	NaN	NaN

[604965 rows x 11 columns]>

```
[10]: # drop None values and save Binned review data as CSV
df_review_data_binned = df_review_data_binned.dropna(subset=['stars_binned']).
    ↪copy()
df_review_data_binned['stars_binned'] = df_review_data_binned['stars_binned'].
    ↪astype(int)
df_review_data_binned.to_csv('/content/drive/My Drive/notebooks/Yelp JSON/
    ↪yelp_dataset (Unzipped Files)/yelp_review_data_binned.csv', index=False)
```

```
[12]: !pip install weasyprint
!pip install ydata-profiling
import pandas as pd
from ydata_profiling import ProfileReport
from weasyprint import HTML

# Generate and display user data profiling report
```

```

review_profile = ProfileReport(df_review_data_binned, title="Profiling Report_
↳Review")
#business_profile.to_notebook_iframe()
review_profile.to_file("/content/drive/My Drive/notebooks/profiling_report.
↳html")
# add sleep
import time
time.sleep(15)

HTML("/content/drive/My Drive/notebooks/profiling_report.html").write_pdf("/
↳content/drive/My Drive/notebooks/profiling_report.pdf")

```

Requirement already satisfied: weasyprint in /usr/local/lib/python3.12/dist-packages (66.0)

Requirement already satisfied: pydyf>=0.11.0 in /usr/local/lib/python3.12/dist-packages (from weasyprint) (0.11.0)

Requirement already satisfied: cffi>=0.6 in /usr/local/lib/python3.12/dist-packages (from weasyprint) (2.0.0)

Requirement already satisfied: tinyhtml5>=2.0.0b1 in /usr/local/lib/python3.12/dist-packages (from weasyprint) (2.0.0)

Requirement already satisfied: tinycss2>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from weasyprint) (1.4.0)

Requirement already satisfied: cssselect2>=0.8.0 in /usr/local/lib/python3.12/dist-packages (from weasyprint) (0.8.0)

Requirement already satisfied: Pyphen>=0.9.1 in /usr/local/lib/python3.12/dist-packages (from weasyprint) (0.17.2)

Requirement already satisfied: Pillow>=9.1.0 in /usr/local/lib/python3.12/dist-packages (from weasyprint) (11.3.0)

Requirement already satisfied: fonttools>=4.0.0 in /usr/local/lib/python3.12/dist-packages (from fonttools[woff]>=4.0.0->weasyprint) (4.60.1)

Requirement already satisfied: pycparser in /usr/local/lib/python3.12/dist-packages (from cffi>=0.6->weasyprint) (2.23)

Requirement already satisfied: webencodings in /usr/local/lib/python3.12/dist-packages (from cssselect2>=0.8.0->weasyprint) (0.5.1)

Requirement already satisfied: brotli>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from fonttools[woff]>=4.0.0->weasyprint) (1.1.0)

Requirement already satisfied: zopfli>=0.1.4 in /usr/local/lib/python3.12/dist-packages (from fonttools[woff]>=4.0.0->weasyprint) (0.4.0)

Requirement already satisfied: ydata-profiling in /usr/local/lib/python3.12/dist-packages (4.17.0)

Requirement already satisfied: scipy<1.16,>=1.4.1 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (1.15.3)

Requirement already satisfied: pandas!=1.4.0,<3.0,>1.1 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (2.2.2)

Requirement already satisfied: matplotlib<=3.10,>=3.5 in

/usr/local/lib/python3.12/dist-packages (from ydata-profiling) (3.10.0)
 Requirement already satisfied: pydantic>=2 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (2.11.10)
 Requirement already satisfied: PyYAML<6.1,>=5.0.0 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (6.0.3)
 Requirement already satisfied: Jinja2<3.2,>=2.11.1 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (3.1.6)
 Requirement already satisfied: visions<0.8.2,>=0.7.5 in /usr/local/lib/python3.12/dist-packages (from visions[type_image_path]<0.8.2,>=0.7.5->ydata-profiling) (0.8.1)
 Requirement already satisfied: numpy<2.2,>=1.16.0 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (2.0.2)
 Requirement already satisfied: minify-html>=0.15.0 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (0.18.1)
 Requirement already satisfied: filetype>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (1.2.0)
 Requirement already satisfied: phik<0.13,>=0.11.1 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (0.12.5)
 Requirement already satisfied: requests<3,>=2.24.0 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (2.32.4)
 Requirement already satisfied: tqdm<5,>=4.48.2 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (4.67.1)
 Requirement already satisfied: seaborn<0.14,>=0.10.1 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (0.13.2)
 Requirement already satisfied: multimethod<2,>=1.4 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (1.12)
 Requirement already satisfied: statsmodels<1,>=0.13.2 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (0.14.5)
 Requirement already satisfied: typeguard<5,>=3 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (4.4.4)
 Requirement already satisfied: imagehash==4.3.1 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (4.3.1)
 Requirement already satisfied: wordcloud>=1.9.3 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (1.9.4)
 Requirement already satisfied: dacite>=1.8 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (1.9.2)
 Requirement already satisfied: numba<=0.61,>=0.56.0 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (0.60.0)
 Requirement already satisfied: PyWavelets in /usr/local/lib/python3.12/dist-packages (from imagehash==4.3.1->ydata-profiling) (1.9.0)
 Requirement already satisfied: pillow in /usr/local/lib/python3.12/dist-packages (from imagehash==4.3.1->ydata-profiling) (11.3.0)
 Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from Jinja2<3.2,>=2.11.1->ydata-profiling) (3.0.3)
 Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib<=3.10,>=3.5->ydata-profiling) (1.3.3)


```

profiling) (2025.10.5)
Requirement already satisfied: patsy>=0.5.6 in /usr/local/lib/python3.12/dist-
packages (from statsmodels<1,>=0.13.2->ydata-profiling) (1.0.2)
Requirement already satisfied: attrs>=19.3.0 in /usr/local/lib/python3.12/dist-
packages (from
visions<0.8.2,>=0.7.5->visions[type_image_path]<0.8.2,>=0.7.5->ydata-profiling)
(25.4.0)
Requirement already satisfied: networkx>=2.4 in /usr/local/lib/python3.12/dist-
packages (from
visions<0.8.2,>=0.7.5->visions[type_image_path]<0.8.2,>=0.7.5->ydata-profiling)
(3.5)
Requirement already satisfied: puremagic in /usr/local/lib/python3.12/dist-
packages (from
visions<0.8.2,>=0.7.5->visions[type_image_path]<0.8.2,>=0.7.5->ydata-profiling)
(1.30)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-
packages (from python-dateutil>=2.7->matplotlib<=3.10,>=3.5->ydata-profiling)
(1.17.0)

Summarize dataset: 0%|          | 0/5 [00:00<?, ?it/s]

```

```

0%|          | 0/11 [00:00<?, ?it/s]
9%|          | 1/11 [00:51<08:33, 51.40s/it]
100%|        | 11/11 [02:18<00:00, 12.55s/it]

```

```

Generate report structure: 0%|          | 0/1 [00:00<?, ?it/s]

```

```

Render HTML: 0%|          | 0/1 [00:00<?, ?it/s]

```

```

Export report to file: 0%|          | 0/1 [00:00<?, ?it/s]

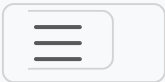
```

Streaming output truncated to the last 5000 lines.

```

WARNING:weasyprint:Ignored `stroke-width: 0.3` at 1:33, unknown property.
WARNING:weasyprint:Ignored `fill: #f0f1f3` at 1:1, unknown property.
WARNING:weasyprint:Ignored `stroke: #f0f1f3` at 1:16, unknown property.
WARNING:weasyprint:Ignored `stroke-width: 0.3` at 1:33, unknown property.
WARNING:weasyprint:Ignored `fill: #f0f1f3` at 1:1, unknown property.
WARNING:weasyprint:Ignored `stroke: #f0f1f3` at 1:16, unknown property.
WARNING:weasyprint:Ignored `stroke-width: 0.3` at 1:33, unknown property.
WARNING:weasyprint:Ignored `fill: #f0f1f3` at 1:1, unknown property.
WARNING:weasyprint:Ignored `stroke: #f0f1f3` at 1:16, unknown property.
WARNING:weasyprint:Ignored `stroke-width: 0.3` at 1:33, unknown property.
WARNING:weasyprint:Ignored `fill: #f0f1f3` at 1:1, unknown property.
WARNING:weasyprint:Ignored `stroke: #f0f1f3` at 1:16, unknown property.
WARNING:weasyprint:Ignored `stroke-width: 0.3` at 1:33, unknown property.
WARNING:weasyprint:Ignored `fill: #0d6efd` at 1:1, unknown property.
WARNING:weasyprint:Ignored `stroke: #0d6efd` at 1:16, unknown property.

```



Overview

Brought to you by [YData](#)

Overview
Alerts **11**
Reproduction

Dataset statistics

Number of variables	11
Number of observations	411589
Missing cells	0
Missing cells (%)	0.0%
Duplicate rows	0
Duplicate rows (%)	0.0%
Total size in memory	53.8 MiB
Average record size in memory	137.1 B

Variable types

Text	4
Categorical	2
Numeric	4
DateTime	1

Variables

review_id
Text

Unique

Distinct	411589
Distinct (%)	100.0%
Missing	0
Missing (%)	0.0%
Memory size	22.4 MiB



More details

user_id

Text

Distinct	199617
Distinct (%)	48.5%
Missing	0
Missing (%)	0.0%
Memory size	22.4 MiB



More details

business_id
Text

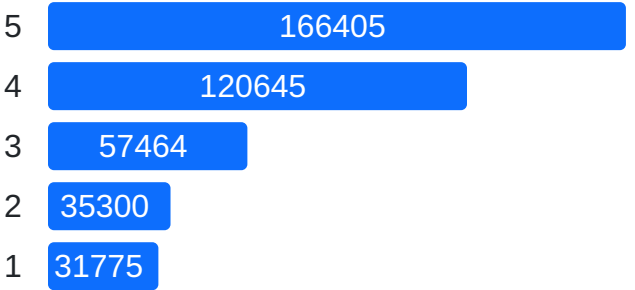
Distinct	31400
Distinct (%)	7.6%
Missing	0
Missing (%)	0.0%
Memory size	22.4 MiB



More details

stars
Categorical

Distinct	5
Distinct (%)	< 0.1%
Missing	0
Missing (%)	0.0%
Memory size	22.4 MiB



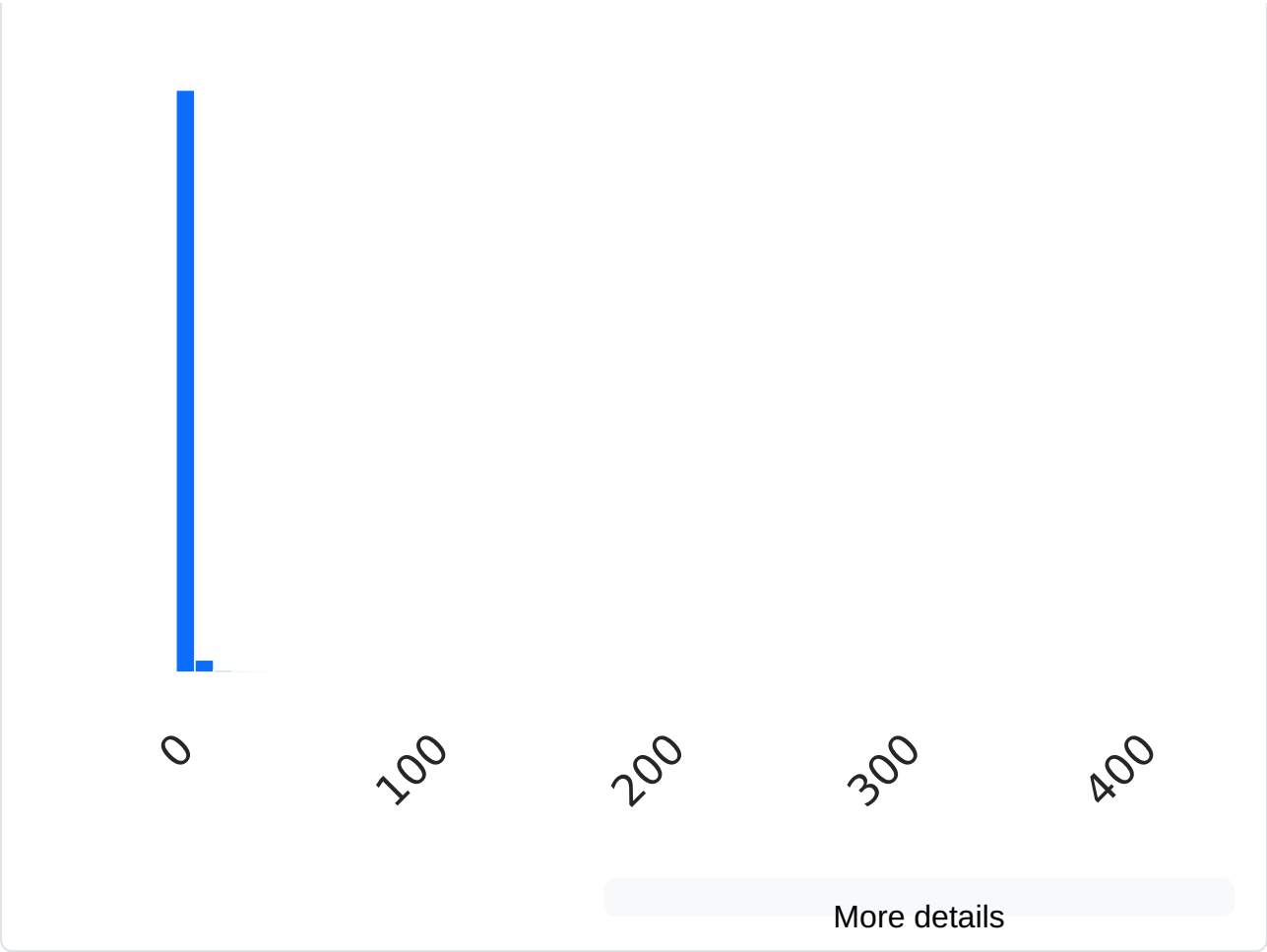
More details

useful

Real number ()

High correlation Skewed Zeros

Distinct	109
Distinct (%)	< 0.1%
Missing	0
Missing (%)	0.0%
Infinite	0
Infinite (%)	0.0%
Mean	1.1634859
Minimum	-1
Maximum	399
Zeros	217200
Zeros (%)	52.8%
Negative	1
Negative (%)	< 0.1%
Memory size	22.4 MiB



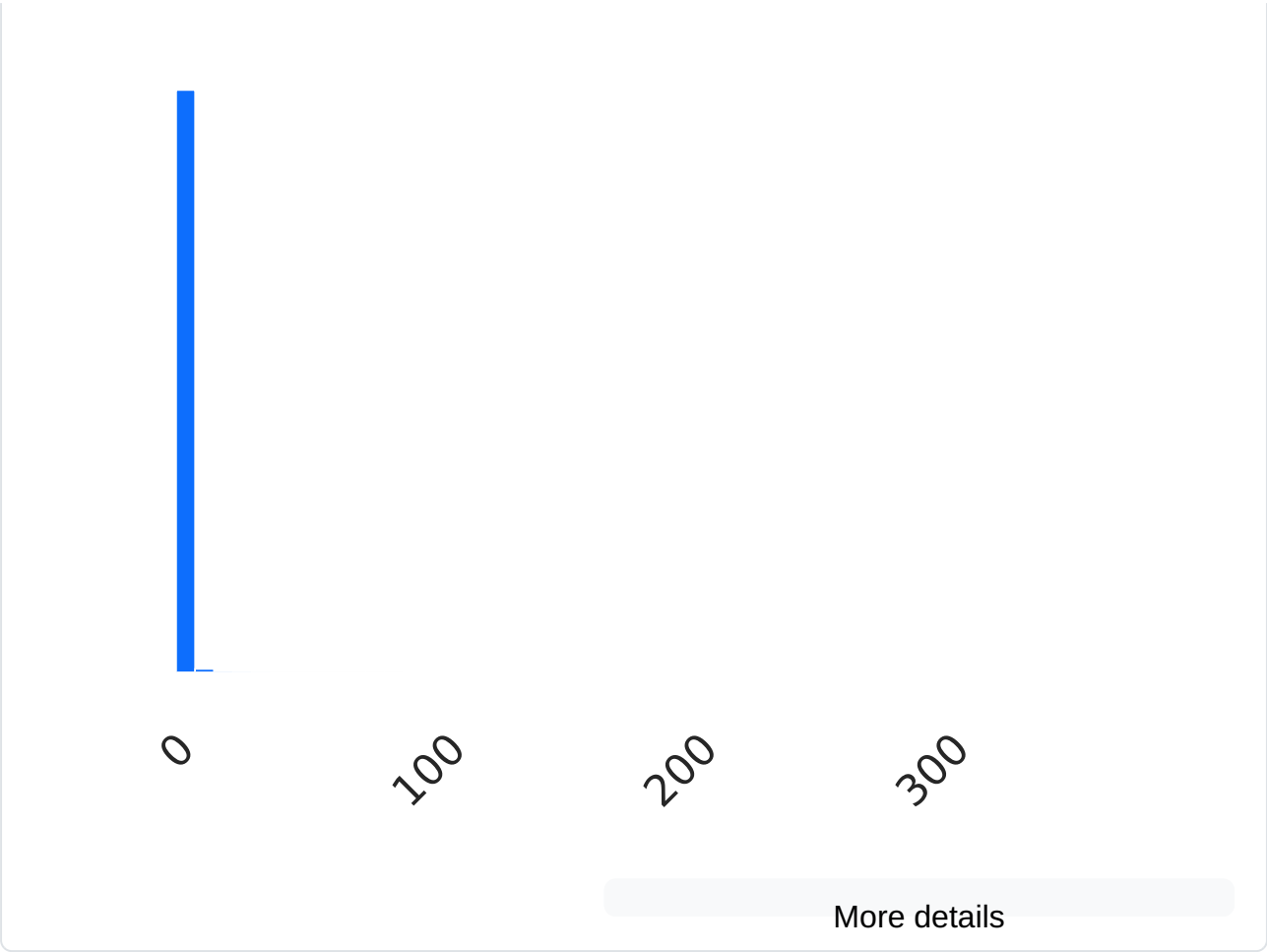
funny

Real number ()

Skewed

Zeros

Distinct	95
Distinct (%)	< 0.1%
Missing	0
Missing (%)	0.0%
Infinite	0
Infinite (%)	0.0%
Mean	0.37395071
Minimum	-1
Maximum	375
Zeros	338460
Zeros (%)	82.2%
Negative	1
Negative (%)	< 0.1%
Memory size	22.4 MiB



cool

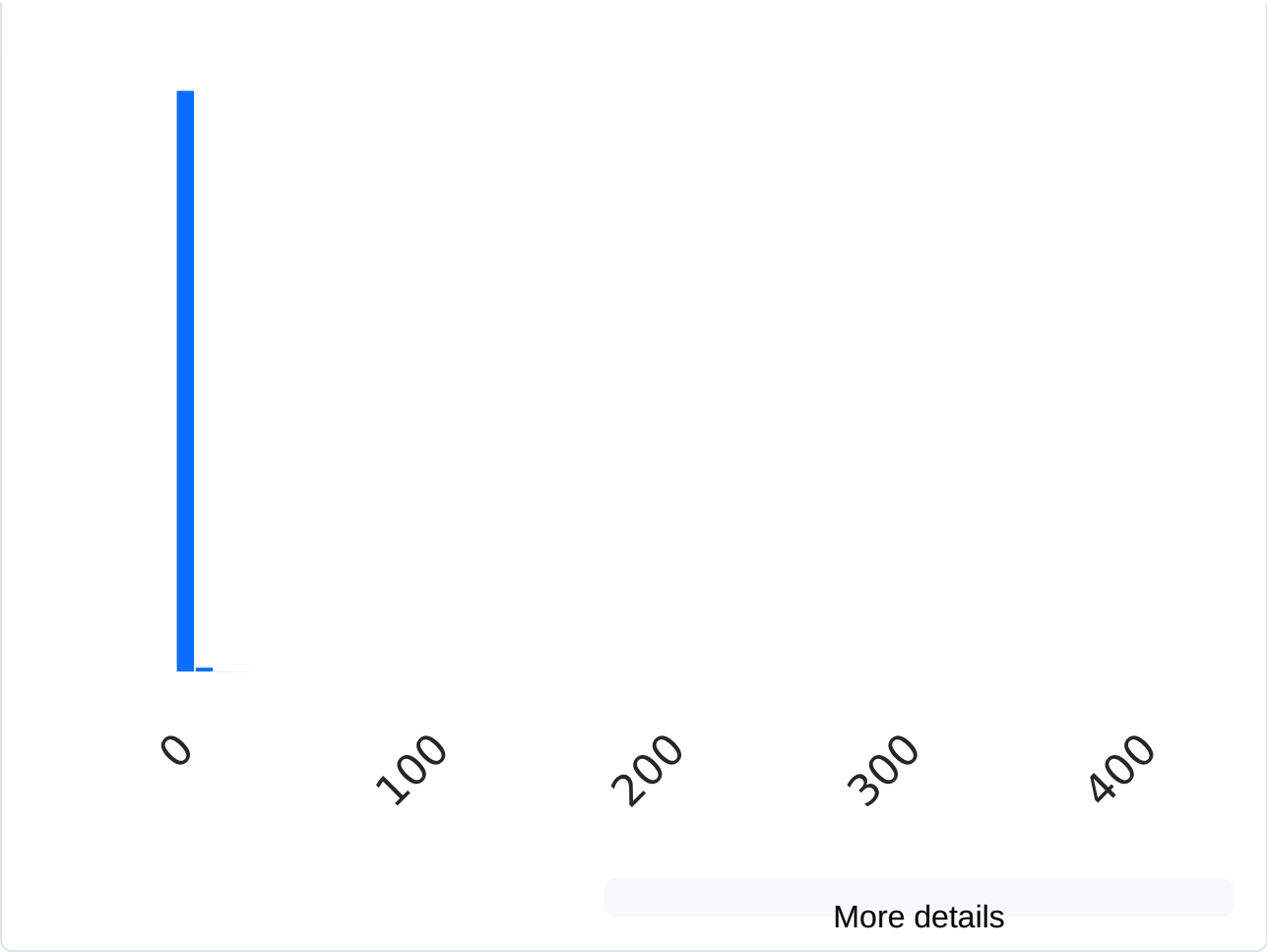
Real number ()

High correlation

Skewed

Zeros

Distinct	101
Distinct (%)	< 0.1%
Missing	0
Missing (%)	0.0%
Infinite	0
Infinite (%)	0.0%
Mean	0.5881231
Minimum	-1
Maximum	399
Zeros	297521
Zeros (%)	72.3%
Negative	1
Negative (%)	< 0.1%
Memory size	22.4 MiB



Text

Distinct	411421
Distinct (%)	> 99.9%
Missing	0
Missing (%)	0.0%
Memory size	22.4 MiB



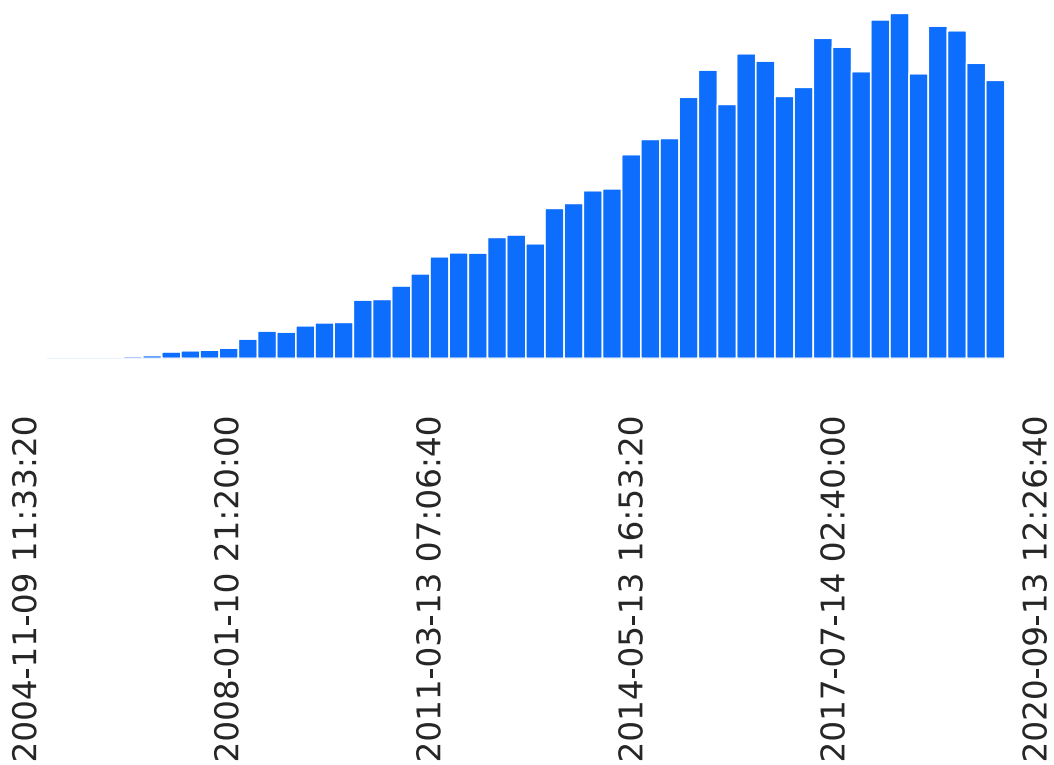
More details

date

Date

Distinct	411204
Distinct (%)	99.9%
Missing	0
Missing (%)	0.0%
Memory size	22.4 MiB

Minimum	2005-03-01 19:33:35
Maximum	2020-03-10 23:49:56
Invalid dates	0
Invalid dates (%)	0.0%



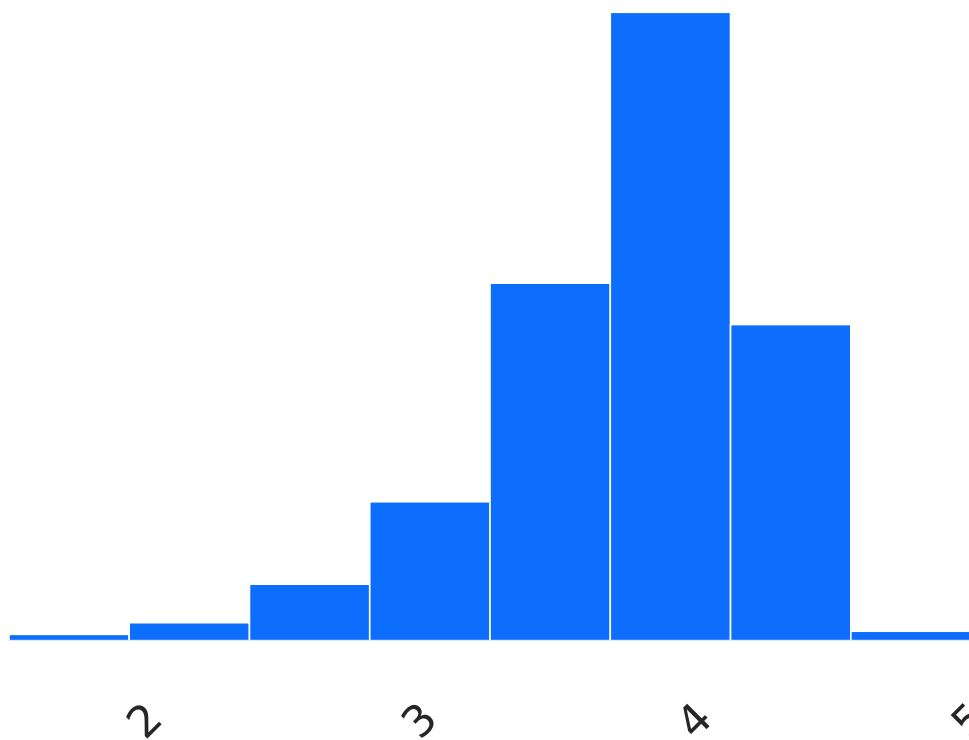
More details

stars_business

Real number ()

High correlation

Distinct	8
Distinct (%)	< 0.1%
Missing	0
Missing (%)	0.0%
Infinite	0
Infinite (%)	0.0%
Mean	3.8128242
Minimum	1.5
Maximum	5
Zeros	0
Zeros (%)	0.0%
Negative	0
Negative (%)	0.0%
Memory size	22.4 MiB



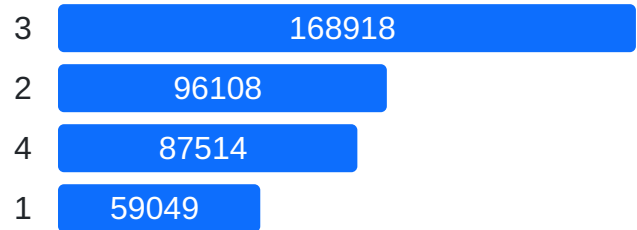
[More details](#)

stars_binned

Categorical

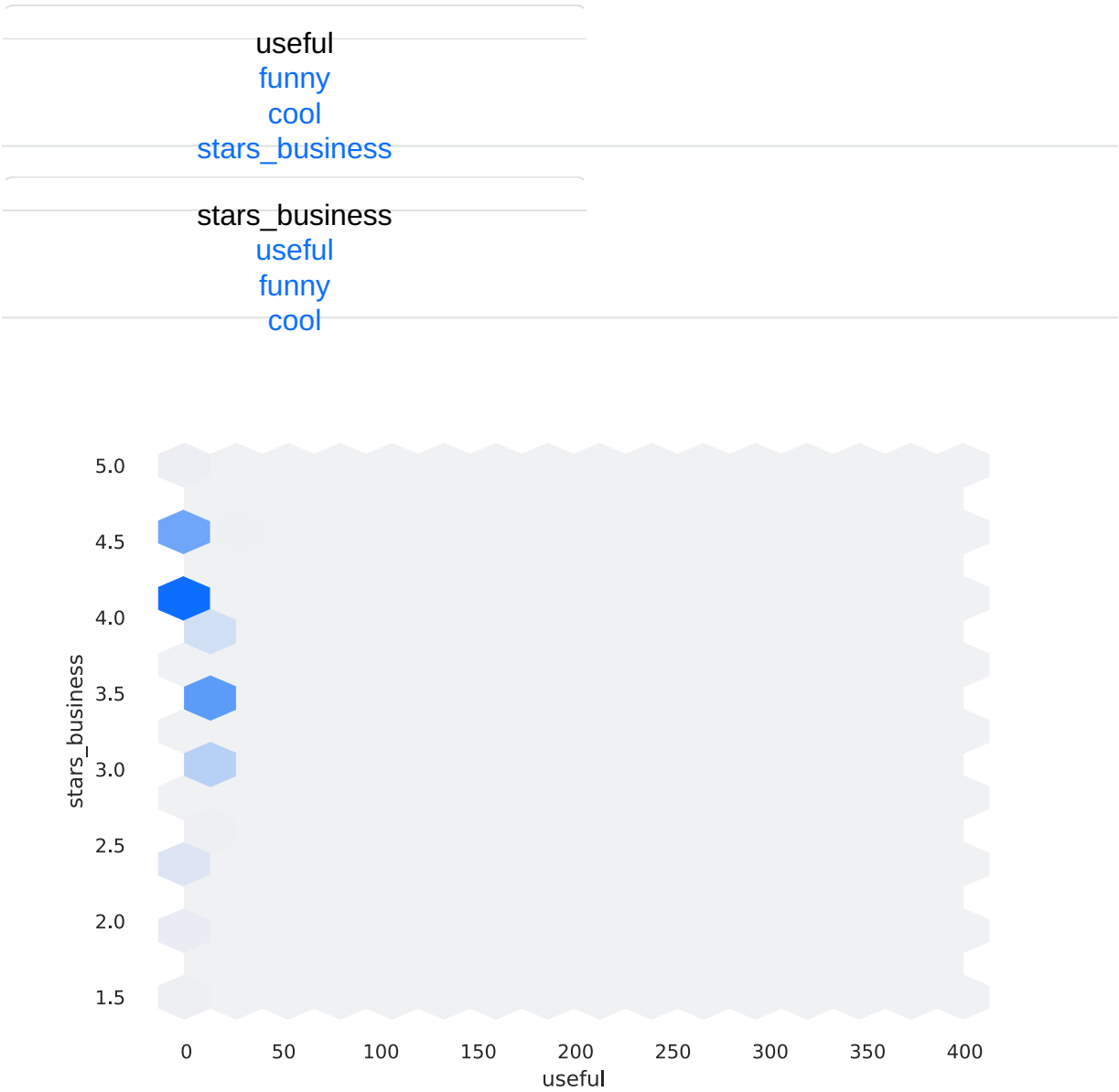
High correlation

Distinct	4
Distinct (%)	< 0.1%
Missing	0
Missing (%)	0.0%
Memory size	22.4 MiB



More details

Interactions

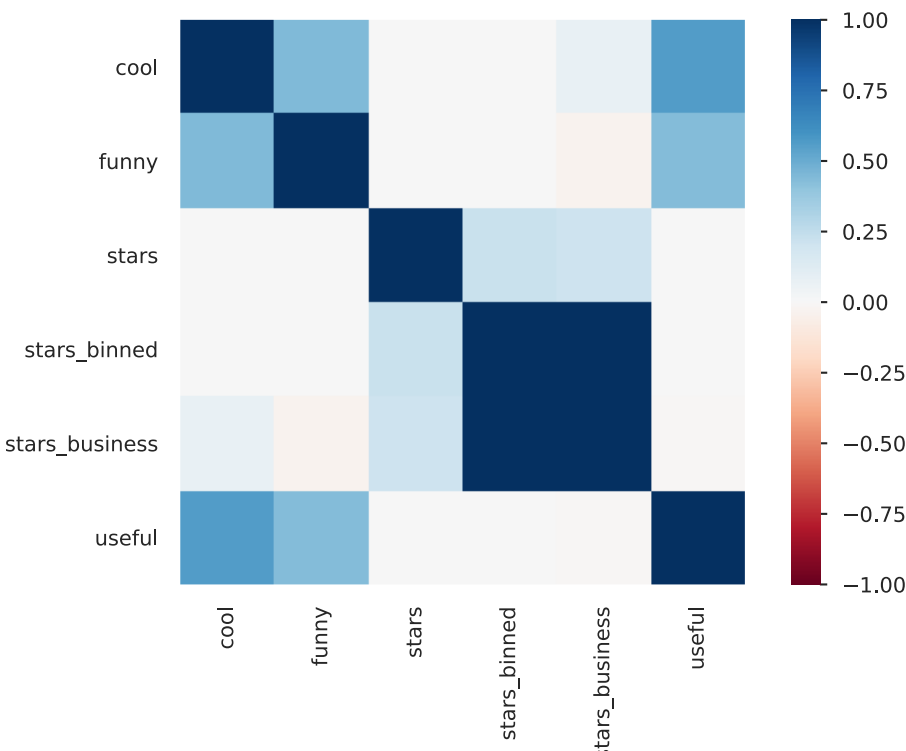


Correlations

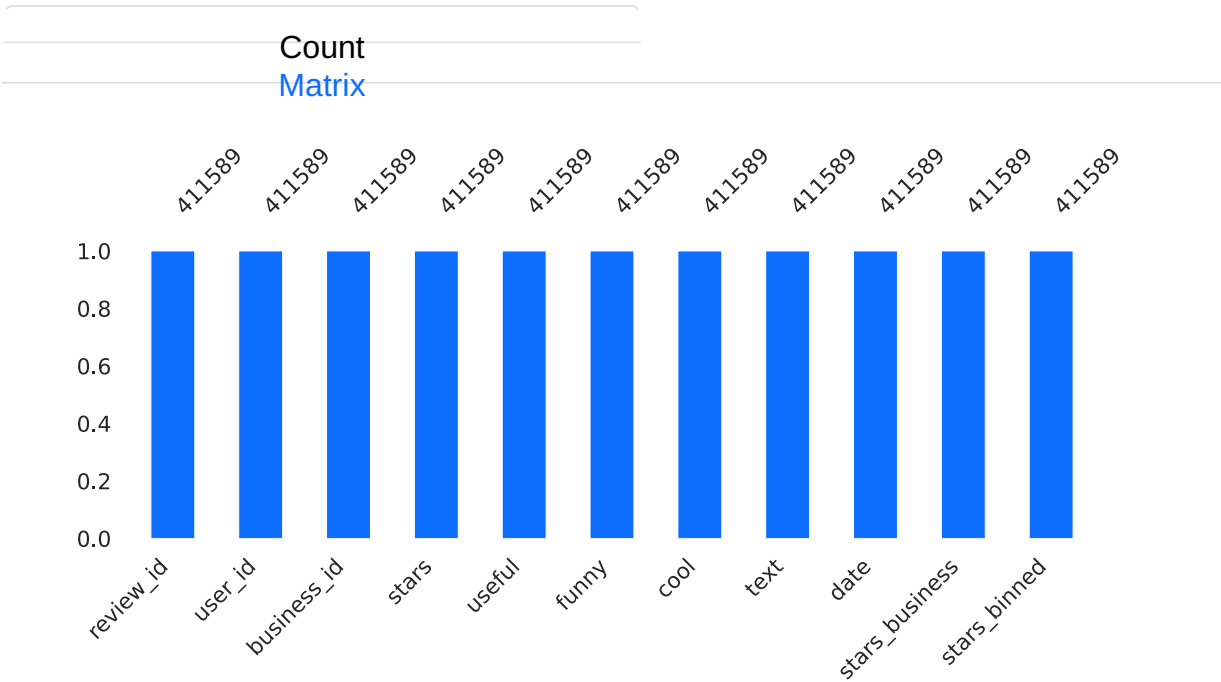
Auto

Heatmap

[Table](#)



Missing values



A simple visualization of nullity by column.

Sample

First rows
[Last rows](#)

	review_id	user_id	business
0	UQLoY-n1QT7roasrWBUudg	vfy68LKvXM7wEHoiUL-IOW	_3m2qyCqx2pnDNyo5vYM3
1	Sn6b49aTOHj_H4qIKA4qwQ	uuJAaJdA_3Dt4CPd-xSU7A	dyVKL3qKsz_XDxtjpbNcgQ
4	GJsPnOd_JgPfMvBF4Xe1Vg	i_40kgr133EgKYVFdBSoZA	8QJTzIaR5FceGds8HBSlig
5	L8Ape_IoEHIMjgiufwCp8Q	c9QdM3JbLT5sCwQ4hGcHIQ	tvbFcYCEQWc76CUK-nBw
6	l84ir9e9ULj7jMt_NcVVxA	4Psac6uj06y_5EPv6CL56g	ju1qrGJ1mcc1sc2oRX7X5g
9	lZzd-DeVV4OKz3vCLEqatw	JzupCYb6JmSOY1uau40ZXw	mY02hlpeZSVCaWObvvyE
11	giW85ug7qQJOorLRpyHGkw	u2oJ1LYgTxHTF3pfJwWnMQ	M0r9lUn2gLFYglwlfG8-bQ
13	nMVoUzD8pbpA73fBmvGjPg	sBn5m-GManXNZjsiM7IHnA	v8lbZ6wcvdyGqWpbDKDvf
16	HXHkEdONfs9nXA9o3e6OMA	rO9ojBDK8lw0x6QVoa4DbA	3ZynJ94VpldDlaArmEp2Rg
17	OtukzIOljacPRI1l5WL7AQ	tGgfOwNHg_sVZZ-DAv7_aA	C_EtrXTygRX5RTUOKtO6D